

Assignment - 16

1) Explain concept of inheritance & what are types of inheritance according to the architecture?

→ Object oriented Paradigms of C++ & Java

- Reusability

- 1 class can acquire char & beh of other class.

Types According to Architecture:

1) Single Inheritance

2) Multi Level Inheritance

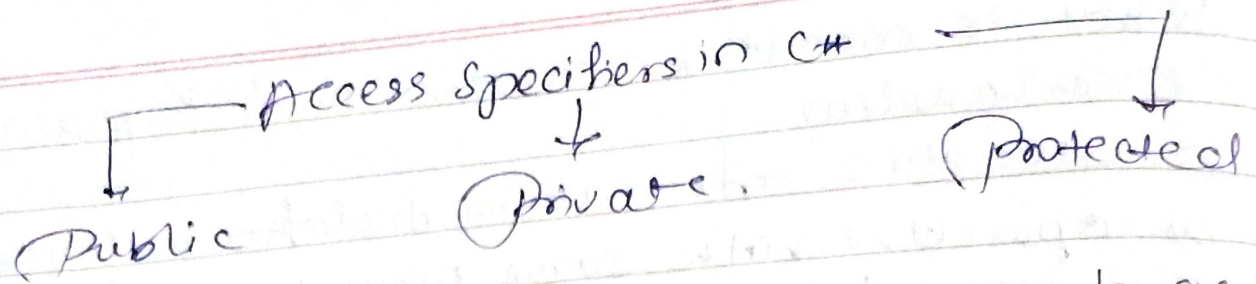
3) Multiple Level Inheritance

4) Hierarchical

5) Hybrid [Combination of 3 types of inheritance]

2) Explain concept of Access Specifier in detail.

→ Indicate who can & who cannot access members (beh + char) of a class



- 1) Public - If we want to allow everyone to access without any restriction
- 2) Private - want to hide something from outside world, only accessible inside the class
- We achieve 'Abstraction'

3) Protected - If class wants to provide access to its child class then that data should be written inside protected access specifier

Codes

1) <u>access.cpp</u>	[public]
2) <u>access1.cpp</u>	[protected]

Q1) What is difference between private & public access specifier
→ Refer Q.2 Code [access.cpp]

Q2) What is default access specifier?
→ A) In C++

1. Inside a class:

default access specifier is private (only within class)

2. Inside a struct:

It is public

B) In Java

Default Access Specifier is called package-private
- If no explicit access modifier is provided, the member is accessible only within the same package.

Q1 What are the types of inheritance according to access specifier

- 1) Public Inheritance - Public & protected members of the base class become public & protected in derived class.
- 2) Protected Inheritance - Public & protected members of base class become protected in derived class.
- 3) Private Inheritance - Public & protected members of base class become private in derived class.

Q2 Explain Constructor & destructor calling sequence in case single level, multi-level & multiple inheritances?

→ Constructor calling sequence

(Top to Bottom)

Single Level &

(Parent)
(DemoC) Top

Multi Level

↓ Bottom
(Hello C) (child)

(Same)

Destructor calling sequence

(Bottom to top)

(Parent)
(DemoC) Top

↓ Bottom
(Hello C) (child)

Multiple Level

Constructor & Destructor calling sequence depends on : Sequence of inheritance.

Q3 When we create object of derived class :

- Top to Bottom
- 1st constructor of base class (Demo) gets called
 - 2nd ———— Parent class (Hello) gets called

7) Draw object layout & class diagram of below code snippets & explain its internal working in detail.
 Explain the type of inheritance in below code snippets.

→

100	<u>bob</u>	Base :: i
104	0	Base :: f
108	0	Base :: d
116	0	

Characteristics :
i, f, d
Behaviour :
fun(), gun()

200	<u>dob</u>	Base :: i
204	0	Base :: f
208	0	Base :: d
216	0	Derived :: i
220	0	Derived :: d
228		

Characteristics :
i, f, d, i, d
Behaviour :
fun(), gun(), gun()

- It is a type of single level inheritance
- When creating an object of Derived (dob):
 - Constructor of base is called 1st to initialize inherited members
 - Constructor of Derived initializes its own member.
- Destructor of Derived - 1st then Base
- Derived :: int i hides base :: int i
- 1 - :: double hides 1 - :: double d.

8) Same Q. as 7.

- The code involves → Multiple inheritance
- Derived class inherits base 1 and base 2
- Object of Derived (dob) is created :

1st - Constructor of base 1	} calling seq.
2nd - Constructor of base 2	
3rd - Constructor of Derived	

100	0	base :: i
104	0	base :: f
108	0	base :: j
112	0	base :: g
116	0	Derived :: i
120	0	Derived :: d
128	0	

Characteristics

i, f, j, g, d

Behaviors

gun(), gun(), sun()

- Derived introduces its own member (int i and double d) hides base :: i

9) ① Same as 7.

- 2 Above is an example of - "Multi Level Inheritance"

100	0	base :: i
104	0	base :: f
108	0	

Characteristics

int i, float f

Behaviors

fun(), gun()

200	0	base :: i
204	0	base :: f
208	0	Derived :: i
212	0	Derived :: d
220	0	

Characteristics

int i, float f, int i, double d

Behaviors

fun(), gun(), sun()

300	0	base :: i
304	0	base :: f
308	0	Derived :: i
312	0	Derived :: d
320	0	Derivedx :: t
324	0	

Characteristics

int i, float f, int i, double d, int k

Behaviors

fun(), gun(), sun(), run()

Here Derived is inheriting base class and further Derivedx is inheriting Derived

Class.
All the characteristics and behaviours of base class are passed to Derived class and further all the characteristics and behaviours of Derived Class are inherited to Derived Class.

10] Same Q. as #]
Above is an example of Hierarchical Inheritance.

16b

100	0	base::i
104	0	base::f
108	0	base::d
116	0	

Characteristics:
int i, float f, double d
Behaviour:
fun(), gun()

24b

200	0	base::i
204	0	base::f
208	0	base::d
216	0	Derived1::x
220	0	Derived1::y
224	0	

Characteristics:
i, f, d, x, y
Behaviours:
fun(), gun(), run()

24b

300	0	
304	0	
308	0	
316	0	Derived2::j
320	0	Derived3::k
324	0	

Characteristics:
i, f, d, j, k
Behaviours:
fun(), gun(), run()

11] Derived1 Class and Derived2 Class are inherited from the same class of base class.